

Chemoprevention of Oral Cancer by a Mixture of Chinese Herbs (Anti-Tumor Mixture B)

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Abstract Anti-tumor B (ATB) is a Chinese herbal mixture of six plants (*Sophora tonkinensis*, *Polygonum bistorta*, *Prunella vulgaris*, *Sonchus brachyotus*, *Dictamnus dasycarpus* and *Dioscorea bulbifera*). ATB is also called Zeng Sheng Ping (ZSP). Previous studies have shown significant chemopreventive efficacy of ATB against human esophageal cancer, mostly squamous cell carcinomas. The chemopreventive effects of ATB on the development of 4-nitroquinoline-1-oxide (4NQO) induced oral squamous cell carcinomas in A/J mice were systematically evaluated and it was found that ATB reduced 4NQO-induced oral cancer development by approximately 60%. Similar results were also seen in 7,12-dimethylbenz(a)anthracene (DMBA) induced oral cancer in hamsters. A randomized clinical trial was conducted in patients with oral leukoplakia. Treatment with ATB (4 tablets, 3 times per day for 8—12 months) reduced the size of oral lesions in 68% of the patients versus 17% of the patients in the placebo control group ($P < 0.01$). These results indicate that ATB may be a potent chemopreventive agent against the development of oral cancer in humans.

Key words antitumor B; Chinese herbal mixture; animal models; oral cancer; oral leukoplakia

1 Introduction

The Chinese herbal mixture, anti-tumor B (ATB), also known as Zeng Sheng Ping (ZSP), is a botanical agent composed of six Chinese herbs: *Sophora tonkinensis*, *Polygonum bistorta*, *Prunella vulgaris*, *Sonchus brachyotus*, *Dictamnus dasycarpus*, and *Dioscorea bulbifera*. ATB is formulated in tablets (Table 1)^[1]. ATB is very effective in the chemoprevention of upper aerodigestive tract tumors in humans and also safe to several thousand subjects over a period of more than two decades^[2–4]. Pre-clinical studies have shown that ATB significantly inhibits precancerous lesions of the bladder^[5]. In this study, the precancerous lesions of the bladder were induced by *N*-butyl-(4-hydroxybutyl) nitrosamine (BBN) in rats. After treatment of rats for 13 months with ATB, the rats

were killed for pathomorphological examination of the bladder. The results showed that the incidence of bladder cancer in the ATB-treated group was reduced by 90.7% ($p < 0.01$) as compared with control rats. This study provided a promising indication of the role of ATB in the prevention of bladder cancer recurrence^[5]. However, there are no follow-ups on the clinical trial in the prevention of bladder cancer recurrence by using

Table 1 Composition of antitumor B

Latin name	Chinese name	form	percentage*
<i>Sophora tonkineusis</i>	Shan Dou Gen	water extract	18%—24%
<i>Polygonum bistorta</i>	Quan Shen	water extract	17%—21%
<i>Prunella vulgaris</i>	Xia Ku Cao	water extract	18%—25%
<i>Sonchus brachyotus</i>	Bei Bai Jiang	water extract	17%—23%
<i>Dictamnus dasycarpus</i>	Bai Xian Pi	water extract	3%—6%
<i>Dioscorea bulbifera</i>	Huang Yao Zi	powder	8%—12%

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ATB from the same group of scientists or others.

The chemopreventive effects of ATB were investigated on chemically induced lung adenocarcinomas using A/J mice harboring a dominant-negative p53 and/or heterozygous deletion of Ink4a/Arf. We found that ATB is a potent lung cancer chemopreventive agent. The inhibitory effects of ATB appear to occur mainly in the tumor progression stage through a p53-, Ink4a/Arf-independent pathway^[6]. In a subsequent study, we tested the chemopreventive effect of ATB on a newly developed lung squamous cell carcinoma (SCC) model^[7]. We demonstrated the chemopreventive effect of ATB on lung SCC as ATB decreased lung SCC significantly (3.1-fold; $p < 0.05$) and increased lung hyperplastic lesions by 2.4-fold ($p < 0.05$). This observation suggests that ATB can block hyperplasia from its progression to SCC. Collectively, our studies indicate that ATB is a potent chemopreventive agent against the development of a majority of non-small cell lung cancer (NSCLC)^[6,7]. Clinical studies showed an approximate 50% reduction in the cancerization rate of marked esophageal dysplasia with ATB^[2–4]. In this clinical trial, more than 2500 cases of marked esophageal dysplasia were randomly divided into groups for ATB and placebo, respectively^[2–4]. After 3 or 5 years of treatment, the progression of esophageal dysplasia to esophageal cancer was inhibited by a remarkable 52.2% and 47.3%, respectively^[2–4].

Head and neck cancer refers to a group of biologically similar cancers originating from the upper aerodigestive tract including the lips, oral cavity, nasal cavity, paranasal sinuses, pharynx, and larynx. Most head and neck cancers are squamous cell carcinomas (SCCs) originating from the flattened epithelium of these regions. The development of oral SCC (OSCC) is positively correlated with certain environmental and lifestyle risk factors such as tobacco smoking, alcohol consumption, and certain strains of sexually transmitted human papillomavirus^[8–12]. Head and neck cancer accounts for about 3%–5% of all malignancies in Western countries with cancer of the oral cavity accounting for 30% of all head and neck cancers. Oral cancer is the sixth most frequent cancer in the world and approximately 36 540 new cases are diagnosed

annually in the United States. Approximately 7 880 patients die from this disease annually^[13]. So far, SCC is the most common malignant neoplasm of the oral cavity. More than 90% of oral cancer is SCC, of which most have premalignant epithelial lesion stages, such as oral leukoplakia and erythroplakia^[14]. The survival rate of patients with SCC of the head and neck (HNSCC) remains poor^[15]. Development of effective chemopreventive interventions of HNSCC is urgently needed.

2 Preclinical studies of ATB as a chemopreventive agent against oral carcinogenesis

2.1 ATB as a chemopreventive agent in 4-nitroquinoline-1-oxide (4NQO) - induced oral SCC in mice

Previous reports by Steidler et al.^[16] followed by Yuan et al.^[17] and our study by Zhang et al.^[18] have shown that 4NQO - induced oral SCC is a pre-clinical model for the study of HNSCCs. Since previous studies by Lin PZ et al.^[2–4] have shown significant chemopreventive efficacy of ATB against SCC of esophagus in human patients with dysplasia^[2–4], we have investigated potential chemopreventive effect of ATB on oral SCC in this mouse model^[19]. As SCC shares similarities with several key features of esophageal SCC such as etiology and histopathology, we hypothesized that ATB is an effective agent for preventing oral SCC development.

To test this hypothesis, we used a 4NQO-induced oral SCC model. At an age of 6–7 weeks, wild-type A/J mice were treated orally with 4NQO (Sigma, N8141) solution (5 mg/mL in propylene glycol, Sigma P4347) using a No. 3 camel-hair brush three times a week for 16 consecutive weeks to induce oral SCCs^[16–18]. Administration of ATB was initiated 8 weeks following the first dose of 4NQO. Powdered ATB tablets were mixed into AIN76A diet in a 3-to-7 ratio and used as the sole food for 24 consecutive weeks. We found that the average tumor number was ~0.5 and the average tumor size was ~10.4 mm³ in control mice, the average tumor number was ~0.2 and the average tumor size was 39.3 mm³ with the ATB treatment, and the oral tumor incidence was decreased from 46% to 19%

(59% decrease). The average tumor number in A/J mice was decreased from 0.5 to 0.2, representing a decrease of ~65%. The decreased oral SCC incidence and decreased oral SCC multiplicity indicated that ATB exerted a positive preventive effect on 4NQO-induced oral tumors in A/J mice. Our results on the effect of ATB on a 4NQO-induced oral SCC model were latterly confirmed by another study reported by Sun Z et al.^[1]. In their study on the 4NQO-induced oral-esophageal cancer model in mice, they found ATB (10% in diet) significantly reduced the incidence of tongue SCC from 55% to 22% ($P < 0.05$) and slightly reduced the incidence of esophageal SCC from 35% to 22%^[1].

Prior to this study, we tested ATB for its chemopreventive efficacy in non-small cell lung cancers in mice^[6, 7]. We found that ATB was well tolerated in A/J mice with doses as high as 300 g/kg diet, and we did not see any dramatic body weight changes in ATB-treated mice (<10% as compared to diet control mice in a given time). Similar body weights were observed in mice in the current study: at the end of the bioassay, ATB-treatment mice had approximately 6.7% lower average body weight as compared to their normal-diet controls. ATB contains hundreds of chemicals as shown from preliminary works of chemical profiling^[20, 21]. We have selected three phytochemicals (obacunone, matrine, and maackiain) and used them as markers for the tissue distribution of ATB. The amounts of these chemicals in the food were $99.6 \pm 10 \mu\text{g}$ of matrine/gram (feed), $83.3 \pm 3.7 \mu\text{g}$ of obacunone/gram, and $6.4 \pm 0.34 \mu\text{g}$ of maackiain/gram. We found significant amounts of all three chemicals in the blood in the ng/mL range after one week of treatment with ATB. The concentrations of these three compounds did not increase further at 18 weeks suggesting that plasma concentrations had reached a steady-state level at one week. The dose of ATB used in the mouse study (300 g/kg diet) is very high because only a small fraction of the matrine (the marker compound) consumed by the mice is absorbed when compared to humans^[7]. Although the amount of matrine or ATB absorbed is small, the absorption does occur quickly and then distributes to the various animal tissues. Lung is an important site of ATB disposition^[7].

Our previous data^[6] showed that ATB can affect

cell proliferation, differentiation, and apoptosis by modulation of the cross-talk among several cellular signaling pathways, especially G13, Notch, and MAPK in the mouse lung tumor model. To investigate possible effects of ATB on cell proliferation and/or apoptosis in oral SCC, we have performed immunohistochemical (IHC) analysis. To investigate the effects of ATB on cell proliferation, we assayed Ki-67 cells in normal, dysplastic, and SCC epithelium of A/J mice. We noticed ~32% ($p = 0.0150$) inhibition of ATB on Ki-67 in dysplasia and a trend towards inhibition in SCC^[19]. The inhibitory effect was associated with a significant decrease of another two biomarkers of cell proliferation: silver stained nucleolar organizer region (AgNOR) and proliferating cell nuclear antigen (PCNA)^[1]. To our surprise, although ATB exhibited a somewhat promoting effect on the induction of apoptosis in 4NQO-induced oral SCC (by IHC staining of BAX and CC3), the effect was not statistically significant^[19]. Epidermal growth factor receptor (EGFR) protein is a member of the erbB family of receptor tyrosine kinases, and it is activated by ligand binding and then undergoes phosphorylation. Activation of EGFR signaling is often associated with cell growth, proliferation, and survival via downstream signaling pathways. Overexpression of EGFR has been documented in a variety of human tumors of epithelial origin including over 80% of oral SCC and SCC of other organs^[22–25]. We have measured the expressions of total EGFR and phosphorylated EGFR at Tyr1173 by IHC staining in 4NQO-treated tongues with or without ATB treatment, and found that the expression of total EGFR and phosphorylated EGFR (Tyr1173) were down-regulated by ATB^[19]. Collectively, these results suggest that ATB is a promising agent for human oral cancer chemoprevention.

2.2 Effect of ATB in 7,12-dimethylbenz[a]anthracene (DMBA)-induced SCC in hamster cheek pouch

Recently, a study was published on chemopreventive efficacy of ATB conducted by Sun et al.^[1] on 7,12-dimethylbenz[a]anthracene (DMBA)-induced SCC in hamster cheek pouch. In their study, six-week old male Syrian golden hamsters were divided into three

groups: control hamsters (no treatment), DMBA treated hamsters (topically treated with 0.5% DMBA three times a week for six weeks), and DMBA+ATB treated-hamsters. The ATB (6 g/kg BW/day) was delivered by oral gavage for 10 weeks. At the end of 16 weeks, they found that DMBA induced about 4 visible tumors per animal with an average of 70 mm³ in size. When compared to the DMBA-treated group, treatment with ATB [1] significantly reduced the incidence of SCC in hamster cheek pouch, which decreased from 95% to 55% ($p < 0.01$). ATB also [1] significantly reduced both tumor number and tumor volume significantly ($p < 0.01$). All hamsters were in good health throughout the experiment without a significant change in body weight. Also, the labeling indexes of two biomarkers of cell proliferation (AgNOR and PCNA) were significantly inhibited by ATB ($p < 0.01$). This study demonstrated that oral intake of ATB had chemopreventive effects on carcinogen-induced oral cancer in hamsters [1]. This observation of ATB on the model of SCC in hamster cheek pouch is further confirmed by the same research group with fractions of ATB applied topically [26].

Guan et al. [26] designed a study to evaluate the therapeutic effects of ATB and extracts of ATB through topical application. They used the same oral SCC model as described by Sun et al. [1]. Based on the diverse chemical structures, they made four different fractions extracted from ATB tablets: *n*-butanol (FB), ethyl acetate (FE), methanol (FM), and water (FW). In their study, Guan et al. [26] applied each of the four fractions locally to DMBA-induced oral cancer lesions after the oral tumor was visualized in golden hamsters three times per week for 18 weeks. They found two out of four fractions (FB and FW) significantly reduced tumor volume by 32.6% ($p < 0.01$) and 22.9% ($p < 0.01$), respectively. By week 24, the average tumor volume was 384.56 mm³ in the DMBA controls. The average tumor volumes in FB or FW treated groups were ~125 mm³ and ~88 mm³, respectively, which were significantly lower as compared to the DMBA controls. However, the treatment with FM or FE showed no significant difference in efficacy. In addition, similar inhibitory effects were seen on the numbers of carcinomatous

lesions in the oral cavity by topical application of FB or FW. Histopathological examinations revealed that an anti-cancer effect from FW may reverse tumorigenesis process.

3 A randomized study of ATB as a chemopreventive agent in patients with oral Leukoplakia

Oral leukoplakia is a type of white patch on the mucous membranes of the oral cavity and tongue with highly variable shapes. Oral leukoplakia is an adherent keratosis which cannot be scraped off easily. However, it is not a specific disease. Leukoplakic lesions are found in approximately 3% of the world's population. Leukoplakia is primarily caused by the use of tobacco. About 5% to 25% of leukoplakias are premalignant lesions; therefore, all leukoplakias should be treated as premalignant lesions by dentists and physicians—they require histologic evaluation or biopsy. Currently, the treatment of leukoplakia mainly involves avoidance of inducing factors such as tobacco smoking. Although several studies have been done on the chemoprevention of oral leukoplakia in patients in clinical trials, only a few agents have shown potential including synthetic retinoid *N*-(4-hydroxyphenyl)-retinamide (4-HPR) [27, 28], alpha-tocopherol (vitamin E) [29], beta-carotene [30], vitamin A [31], Bowman-Birk inhibitor (a protease inhibitor derived from soybeans) [32], black tea [33], and freeze-dried black raspberry gel [34]. However, the promising potential exhibited by these agents needs further confirmation.

Since oral SCC is very similar to esophageal SCC in etiology, histopathology and molecular mechanism, it is likely that ATB may be effective for oral cancer chemoprevention. A recent study by Sun et al. [1] examined the chemopreventive effects of ATB in a short-term clinical trial of oral leukoplakia which is a pre-malignant lesion of oral cancer. In a three year period, they recruited 120 patients with oral leukoplakia in Beijing, China [1]. In this clinical trial, Sun et al. gave ATB tablets or placebo tablets to oral leukoplakia patients. The treatment regimen was 4 tablets each time, 3 times per day for 8–12

months. In the entire trial, they conducted clinical evaluations at the study entry-time, monthly during the intervention, and three months after the study period; they recorded oral leukoplakia lesions with color photos and measured sizes. They defined a positive response as a disappearance or reduction of size by more than 50% at the final checkup, stable disease as an insignificant change in the size, and progressive disease as an increase in the size of the lesion by >50% or the development of new lesions. In the end, they found that the ATB intervention reduced the size of oral leukoplakia in 40 out of 59 patients whereas the placebo was effective in 9 out of 53 patients. The difference is significant ($p < 0.01$). Such an effect was associated with a significant decrease in AgNOR and PCNA-labeling index.

4 Potential molecular mechanisms

It is clear that ATB is an effective preventive agent in cancer development. The logical hypothesis is that ATB functions through a decrease in cell proliferation, an increase in cell apoptosis, or both. Commonly used proliferative markers include Ki-67, BrdU (5-bromo-deoxyuridine), EGFR, proliferating cell nuclear antigen (PCNA), and silver stained nucleolar organizer region (AgNOR), etc. EGFR is activated by ligand binding and then undergoes phosphorylation. Activated EGFR stimulates cell growth, proliferation and survival via downstream signaling pathways. Commonly used apoptotic markers include cleaved caspase 3 (CC3) and Bcl-2 family proteins (Bcl-2, Bcl-x, Bax, Bak, and Bad), etc.

Component analysis of ATB has shown that phytosterols (PS), flavones, alkaloids, sapogenins, triterpenes, and triterpenoids are the major effective chemicals which are known to have antimutagenic, anti-inflammatory, and differentiation effects in cell cultures derived from different types of malignancy^[19,35–38]. Functional analysis revealed that these compounds can stimulate apoptosis and inhibit cell growth^[39–41]. One mechanism for their actions is that the induction of cell cycle arrest at the G2/M phase in breast tumor MDA-MB-231 cells^[42].

To identify specific genes or pathways that may

contribute to ATB's protective effect, altered gene expression in lung tumors treated with ATB was investigated. Oligonucleotide microarray analysis revealed 284 genes that were differentially expressed in mouse lung tumors as compared to the normal lung in wild-type A/J mice. We found that 114 out of these 284 genes were expressed similar to the normal levels in tumors treated with ATB. Most of the genes modulated by ATB belong to several cellular signaling pathways, including Notch, growth factors, G protein-Ras-MAPK, ubiquitin-proteasome (CDC34, Cullin1, 26S proteasome), and apoptosis (BAD promoter, caspase 3)^[6].

More recently, studies have shown that ATB has an effect on oral tumor cells via decreasing cell proliferation. We assayed Ki-67 labelling in the cells of normal, dysplastic and SCC epithelium of mice. We found that ATB inhibited Ki-67 labelling index in dysplasia (31.5% inhibition, $p=0.0150$) and trended towards inhibition in SCC (23.1% inhibition, $p=0.0915$). We also found that the expression of EGFR was decreased by 30.66% ($p=0.0537$, one-tail t test) in mice that received ATB treatment^[19]. In another study, Sun et al.^[1] demonstrated the preventive effect of ATB in three oral lesion models: DMBA-induced oral SCC in hamster cheek pouch, 4NQO-induced oral cancer in mice, and randomized placebo-controlled clinical trial on oral leukoplakia^[1]. They showed that this effect was associated with a significant decrease of cell proliferation by measuring the labeling index of AgNOR and PCNA^[1]. Lately, Guan et al.^[26] showed that ATB inhibits tumor cell proliferation by two out of the four fractions of ATB (*n*-butanol and water fractions). Topical application of ATB markedly decreased the BrdU-positive cell numbers in oral tumor lesions and reduced the expression level of MAPK^[26].

To investigate the effects of ATB on apoptosis in oral SCC cells, we measured altered expression of mediators of apoptosis by immunohistochemical (IHC) staining. We found that, although ATB exhibited a modest promotional effect on the induction of apoptosis in 4NQO-induced oral SCC, the effect was not statistically significant^[19]. Lately, Guan et al.^[26] showed that ATB promotes apoptosis in tumor cells. They found that two out of the four fractions of ATB

(*n*-butanol and water fractions) have therapeutic effects when tested topically in the DMBA-induced oral SCC in hamster cheek pouch. Topical application of ATB markedly increases caspase-3 expression and decreases Bcl-2 protein production^[26].

5 Conclusion and outlook

These studies in pre-clinical models of mice and hamsters provide solid evidence that ATB can effectively prevent oral cancer^[1, 19, 26]. In traditional Chinese medicine, ATB was used in the treatment of precancerous lesions of esophageal cancer. In the clinical trial, Sun et al. obtained promising results of using ATB in the treatment of oral leukoplakia in clinical practice^[1]. This promising trend needs further confirmation before recommending the use of ATB clinically in oral cancer patients.

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增生平用于口腔癌化学预防

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摘 要 增生平 (ZSP, 又称 anti-tumor B, ATB) 是一种由山豆根、拳参、北败酱、夏枯草、白鲜皮、黄药子等组成的中药复方。以前的研究显示增生平对人食管癌 (主要是鳞状细胞癌) 有显著的化学预防效用。我们系统地研究评价了增生平对 4-硝基喹啉-1-氧化物诱导的 A/J 小鼠口腔癌发生发展的影响, 发现增生平减少了约 60% 的口腔癌。对 7,12-二甲苯并 (a) 蒽诱导的仓鼠口腔癌也观察到了类似的结果。进而在口腔白斑患者中开展了一个随机临床实验, 结果发现增生平 (每日三次, 每次 4 片, 持续 8—12 个月) 在 68% 的患者中减少了口腔病变的大小, 而在安慰剂组只有 17% ($P < 0.01$)。这些结果显示增生平可能是一种强效的人口腔癌化学预防试剂。

关键词 增生平 中药复方 动物模型 口腔癌 口腔白斑

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